H524 Introduction to Biostatistics
Week 5 Lab: Hypothesis Testing
Fall 2025

Format: Online
Tools Required: R, RStudio
Estimated Time: 90 minutes

Problem 1: Two-Sided t-Test (Body Temperature)

Research Question: Is the true mean body temperature different from the commonly cited value of 98.6°F?

Data: Body temperatures from $n = 15$ healthy adults:

```
temps <- c(98.4, 98.0, 98.2, 98.6, 98.5, 98.3, 98.1, 98.7,
           98.2, 98.4, 98.3, 98.0, 98.5, 98.4, 98.2)
```

Part (a): State the hypotheses

Null hypothesis:

Alternative hypothesis:

Note: Is this a one-sided or two-sided test? Why?

Part (b): Calculate descriptive statistics

R code:

```
n <- length(temps)
xbar <- mean(temps)
s <- sd(temps)

n
xbar
s
```

Results:

- Sample size: $n =$
- Sample mean: $\bar{x} =$

- Sample standard deviation: $s =$

Part (c): Conduct the hypothesis test

Step 1: Calculate the standard error

Step 2: Calculate the t-statistic

Step 3: Determine degrees of freedom

Step 4: Calculate the p-value

For a two-sided test, we calculate the probability in both tails:

R verification:

```
# Manual calculation
SE <- s / sqrt(n)
t_stat <- (xbar - 98.6) / SE
p_value <- 2 * pt(abs(t_stat), df = n - 1, lower.tail = FALSE)

SE
t_stat
p_value

# Using t.test function
result <- t.test(temps, mu = 98.6, alternative = "two.sided")
result
```

Part (d): Make a decision and interpret

Decision at $\alpha = 0.05$:

Interpretation:

95% Confidence Interval:

Problem 2: One-Sided t-Test (Blood Pressure)

Research Question: Does a new blood pressure medication lower mean systolic blood pressure below 140 mmHg?

Data: Systolic blood pressure from $n = 20$ patients after treatment:

```
bp <- c(138, 135, 142, 128, 145, 137, 132, 140, 136, 134,
        139, 141, 133, 130, 143, 135, 138, 142, 136, 139)
```

Part (a): State the hypotheses

Null hypothesis:

Alternative hypothesis:

Note: Is this a one-sided or two-sided test? Why?

Part (b): Calculate descriptive statistics

R code:

```
n <- length(bp)
xbar <- mean(bp)
s <- sd(bp)

n
xbar
s
```

Results:

- Sample size: $n =$
- Sample mean: $\bar{x} =$
- Sample standard deviation: $s =$

Part (c): Conduct the hypothesis test

Step 1: Calculate the standard error

Step 2: Calculate the t-statistic

Step 3: Determine degrees of freedom

Step 4: Calculate the p-value (one-sided, lower tail)

R verification:

```
# Manual calculation
SE <- s / sqrt(n)
t_stat <- (xbar - 140) / SE
p_value <- pt(t_stat, df = n - 1)

SE
t_stat
p_value

# Using t.test function (one-sided)
result <- t.test(bp, mu = 140, alternative = "less")
result
```

Part (d): Make a decision and interpret

Decision at $\alpha = 0.05$:

Interpretation:

Clinical significance:

Problem 3: Type I and Type II Errors

Using the blood pressure example from Problem 2, explain the concepts of Type I and Type II errors.

Type I Error (α)

Definition:

In this context:

Probability:

Consequence:

Control:

Type II Error (β)

Definition:

In this context:

Probability:

Consequence:

Control:

Statistical Power

Definition:

Interpretation:

Typical goal:

Trade-off

There is an inverse relationship between Type I and Type II errors. Explain:

Problem 4: Power Analysis

Understanding statistical power and sample size determination.

Part (a): Calculate power for a given sample size

Scenario: We want to detect a medium effect (Cohen's $d = 0.5$) with $n = 30$ subjects at $\alpha = 0.05$ (two-sided test).

R code:

```
library(pwr)

power_result <- pwr.t.test(n = 30,
                           d = 0.5,
                           sig.level = 0.05,
                           type = "one.sample",
                           alternative = "two.sided")
power_result
```

Result: Power $\approx$

Interpretation:

Assessment:

Part (b): Calculate sample size for desired power

Question: How many subjects do we need to achieve 80% power for detecting a medium effect ($d = 0.5$)?

R code:

```
n_result <- pwr.t.test(d = 0.5,
                       power = 0.80,
                       sig.level = 0.05,
                       type = "one.sample",
                       alternative = "two.sided")
n_result
```

Result: Required $n =$

Interpretation:

Part (c): Factors affecting power

Power increases when:

Key insight:

Problem 5: Relationship Between Confidence Intervals and Hypothesis Tests

There is a fundamental equivalence between confidence intervals and hypothesis tests.

Equivalence Principle

For a two-sided test at significance level α:

If the $(1-\alpha) \times 100\%$ confidence interval contains the null value μ_0:

If the $(1-\alpha) \times 100\%$ confidence interval does NOT contain μ_0:

Demonstration with Body Temperature Data

Using the body temperature data from Problem 1:

95% Confidence Interval:

Null value: $\mu_0 = 98.6$ °F
Question: Is 98.6 inside the interval?

Answer:

Conclusion from CI:

Conclusion from hypothesis test:

Result: Do both methods agree?

Advantages of Confidence Intervals

Confidence intervals provide MORE information than hypothesis tests:

Recommendation:

Problem 6: Checking the Normality Assumption

The t-test assumes that the data come from a normally distributed population. We should check this assumption.

Methods for Checking Normality

1. Visual Methods:

2. Formal Test:

Applying to Body Temperature Data

R code for diagnostic plots:

```
# Create diagnostic plots
par(mfrow = c(1, 3))

# Histogram
hist(temps, breaks = 5, main = "Histogram of Body Temps",
     xlab = "Temperature (F)", col = "lightblue", border = "white")

# Boxplot
boxplot(temps, main = "Boxplot", ylab = "Temperature (F)")

# Q-Q plot (most important for normality)
qqnorm(temps, main = "Q-Q Plot", pch = 19, col = "blue")
qqline(temps, col = "red", lwd = 2)

par(mfrow = c(1, 1))
```

Shapiro-Wilk Test

R code:

```
shapiro.test(temps)
```

Hypotheses:

- H_0:
- H_A:

Results:

- Test statistic: $W =$
- P-value: $p =$

Decision:

Interpretation:

Assessment

Visual assessment:

Formal test:

Conclusion:

What if normality is violated?

If normality is violated:

Summary of Key Concepts

Steps for Hypothesis Testing

1. **State hypotheses** —
2. **Choose significance level** —
3. **Check assumptions** —
4. **Calculate test statistic** —
5. **Find p-value** —
6. **Make decision** —
7. **Interpret in context** —

One-Sample t-Test Formula

Test statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad \text{with } df = n - 1$$

Two-sided p-value:

$$p = 2 \times P(T \geq |t|)$$

One-sided p-value (lower tail):

$$p = P(T \leq t)$$

One-sided p-value (upper tail):

$$p = P(T \geq t)$$

Error Types

	H_0 **True**	H_0 **False**
Reject H_0		
Fail to Reject H_0		

Important Relationships

- **CI and hypothesis test:**
- **Sample size and power:**
- **Effect size and power:**
- **Alpha and power:**